

ADV: Functions (Adv), F2 Graphing (Adv)
Transformations (Y12)
Non-Calculus Graphing (Y12)

Teacher: Ms Najem

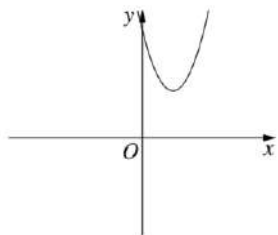
Exam Equivalent Time: 103.5 minutes (based on HSC allocation of 1.5 minutes approx. per mark)

Questions

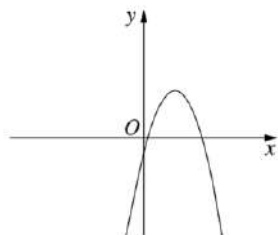
1. Functions, 2ADV F2 2016 HSC 3 MC

Which diagram best shows the graph of the parabola $y = 3 - (x - 2)^2$?

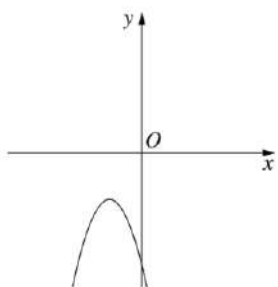
(A)



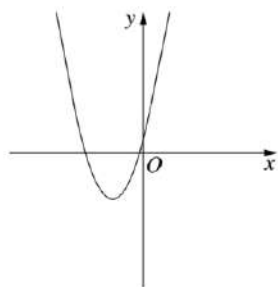
(B)



(C)



(D)

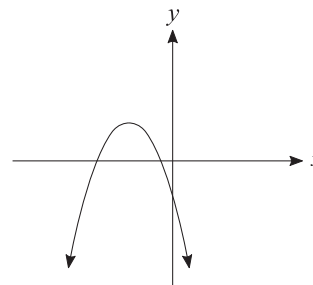


2. Functions, 2ADV F2 EQ-Bank 2 MC

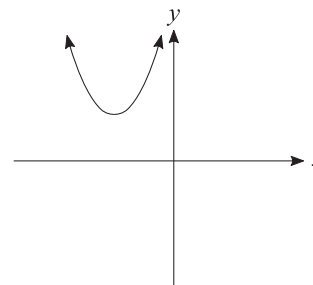
Which diagram best shows the graph

$$y = 1 - 2(x + 1)^2$$

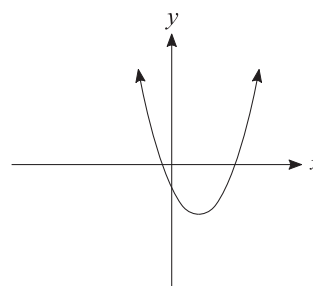
A.



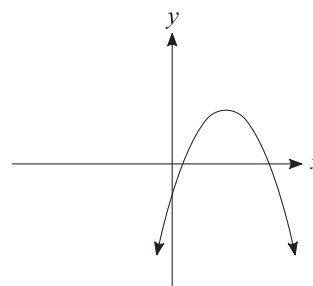
B.



C.

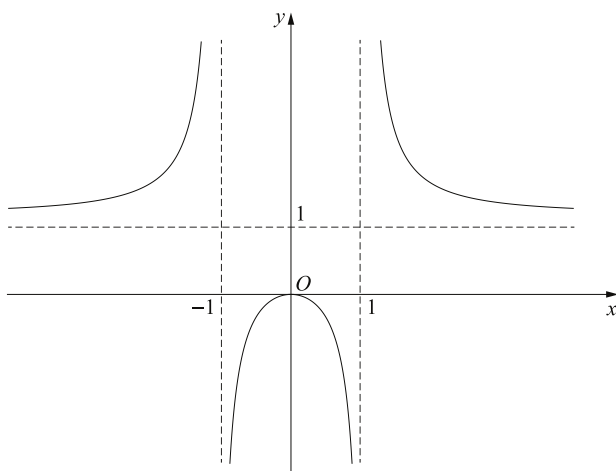


D.



3. Functions, 2ADV' F2 2019 HSC 4 MC

The diagram shows the graph of $y = f(x)$.



Which equation best describes the graph?

- A. $y = \frac{x}{x^2 - 1}$
- B. $y = \frac{x^2}{x^2 - 1}$
- C. $y = \frac{x}{1 - x^2}$
- D. $y = \frac{x^2}{1 - x^2}$

4. Functions, 2ADV F2 SM-Bank 5 MC

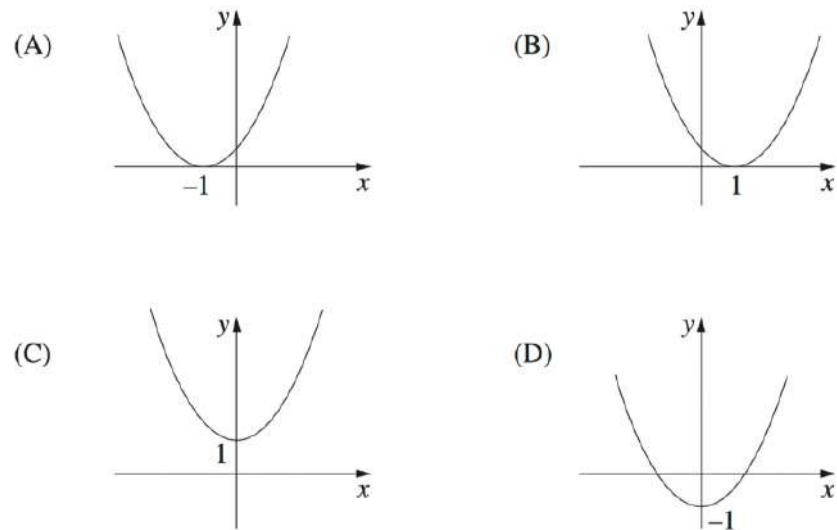
The point $P(4, -3)$ lies on the graph of a function $f(x)$. The graph of $f(x)$ is translated four units vertically up and then reflected in the y -axis.

The coordinates of the final image of P are

- A. $(-4, 1)$
- B. $(-4, 3)$
- C. $(0, -3)$
- D. $(4, -6)$

5. Functions, 2ADV F2 2014 HSC 2 MC

Which graph best represents $y = (x - 1)^2$?



6. Functions, 2ADV F2 SM-Bank 6 MC

The graph of a function $f(x)$ is obtained from the graph of the function $g(x) = \sqrt{2x - 5}$ by a reflection in the x -axis followed by a dilation from the y -axis by a factor of $\frac{1}{2}$.

Which one of the following is the function $f(x)$?

- A. $f(x) = \sqrt{5 - 4x}$
- B. $f(x) = -\sqrt{x - 5}$
- C. $f(x) = \sqrt{x + 5}$
- D. $f(x) = -\sqrt{4x - 5}$

7. Functions, 2ADV F2 SM-Bank 4 MC

The graph of the function $f(x) = 3x^{\frac{5}{2}}$ is reflected in the x -axis and then translated 3 units to the right and 4 units down.

The equation of the new graph is

- A. $y = 3(x - 3)^{\frac{5}{2}} + 4$
- B. $y = -3(x - 3)^{\frac{5}{2}} - 4$
- C. $y = -3(x + 3)^{\frac{5}{2}} - 1$
- D. $y = -3(x - 4)^{\frac{5}{2}} + 3$

8. Functions, 2ADV F2 SM-Bank 10 MC

The graph of the function $f(x) = \frac{x - 3}{2 - x}$ has asymptotes

- A. $x = 3$, $y = 2$
- B. $x = -2$, $y = 1$
- C. $x = 2$, $y = -1$
- D. $x = 2$, $y = 1$

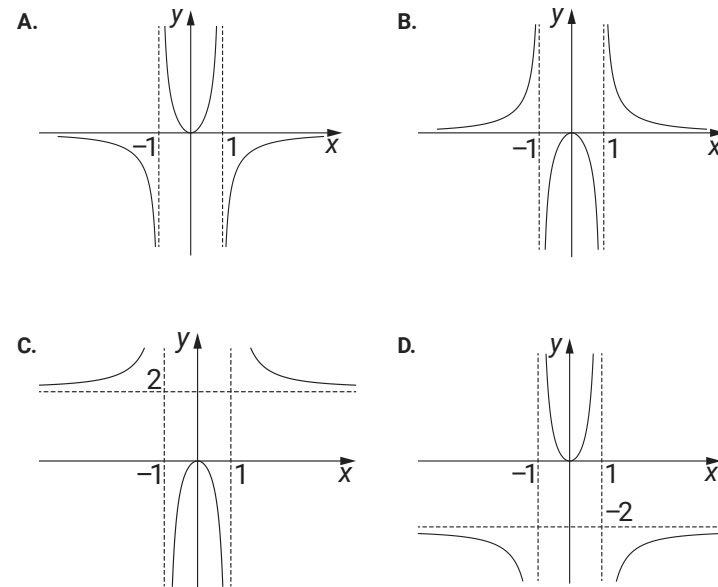
9. Functions, 2ADV' F2 2015 HSC 5 MC

What are the asymptotes of $y = \frac{3x}{(x + 1)(x + 2)}$

- (A) $y = 0$, $x = -1$, $x = -2$
- (B) $y = 0$, $x = 1$, $x = 2$
- (C) $y = 3$, $x = -1$, $x = -2$
- (D) $y = 3$, $x = 1$, $x = 2$

10. Functions, 2ADV' F2 2017 HSC 5 MC

Which graph best represents the function $y = \frac{2x^2}{1 - x^2}$?



11. Functions, 2ADV F2 SM-Bank 7 MC

The point $A(3, 2)$ lies on the graph of the function $f(x)$. A transformation maps the graph of $f(x)$ to the graph of $g(x)$,

where $g(x) = \frac{1}{2}f(x - 1)$. The same transformation maps the point A to the point P .

The coordinates of the point P are

- A. $(2, 1)$
- B. $(2, 4)$
- C. $(4, 1)$
- D. $(4, 2)$

12. Functions, 2ADV F2 SM-Bank 8 MC

The transformation that maps the graph of $y = \sqrt{8x^3 + 1}$ onto the graph of $y = \sqrt{x^3 + 1}$ is a

- A. dilation by a factor of **2** from the **y**-axis.
- B. dilation by a factor of **2** from the **x**-axis.
- C. dilation by a factor of $\frac{1}{2}$ from the **x**-axis.
- D. dilation by a factor of $\frac{1}{2}$ from the **y**-axis.

13. Functions, 2ADV F2 EQ-Bank 5

The function $f(x) = \sin x$ is transformed into the function $g(x) = \frac{\sin(4x)}{3}$.

Describe in words how the amplitude and period have changed in this transformation. (2 marks)

14. Functions, 2ADV F2 2006 HSC 1c

Sketch the graph of $y = |x + 4|$. (2 marks)

15. Functions, 2ADV F2 SM-Bank 1

- i. Draw the graph $y = \ln x$. (1 mark)
- ii. Explain how the above graph can be transformed to produce the graph

$$y = 3\ln(x + 2)$$

and sketch the graph, clearly identifying all intercepts. (3 marks)

16. Functions, 2ADV F2 EQ-Bank 9

$y = -\frac{(x+2)^4}{3}$ has been produced by three successive transformations: a translation, a dilation and then a reflection.

- i. Describe each transformation and state the equation of the graph after each transformation. (2 marks)
- ii. Sketch the graph. (1 mark)

17. Functions, 2ADV F2 EQ-Bank 9

Consider the function $f(x) = \frac{1}{4x-1}$.

- (i) Find the domain of $f(x)$. (1 mark)
- (ii) Sketch $f(x)$, showing all asymptotes and intercepts? (2 marks)

18. Functions, 2ADV' F2 2012 HSC 13b

- (i) Find the horizontal asymptote of the graph

$$y = \frac{2x^2}{x^2 + 9}. \quad (1 \text{ mark})$$

- (ii) Without the use of calculus, sketch the graph

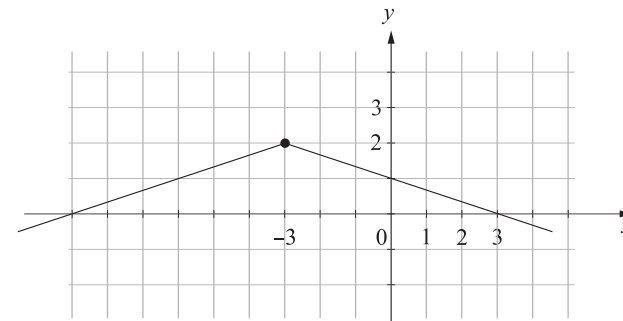
$$y = \frac{2x^2}{x^2 + 9},$$

showing the asymptote found in part (i). (2 marks)

19. Functions, 2ADV F2 EQ-Bank 1

The function $f(x) = |x|$ is transformed and the equation of the new function is $y = kf(x+b)+c$.

The graph of the new function is shown below.



What are the values of k , b and c . (2 marks)

20. Functions, 2ADV F2 EQ-Bank 14

List a set of transformations that, when applied in order, would transform $y = x^2$ to the graph with equation $y = 1 - 6x - x^2$. (3 marks)

21. Functions, 2ADV F2 SM-Bank 13

i. Show that the function $y = \frac{1 - e^x}{1 + e^x}$ is an odd function? (1 mark)

ii. Sketch $y = \frac{1 - e^x}{1 + e^x}$, labelling all intercepts and asymptotes. (2 marks)

22. Functions, 2ADV F2 SM-Bank 2

Sketch the graph $y = \log_2(x - 3)$.

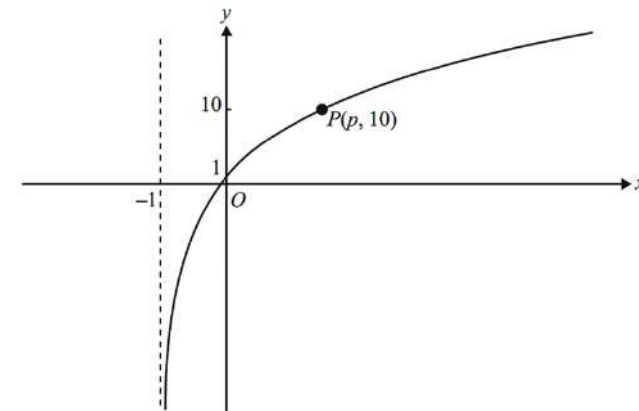
Show all asymptotes and state its domain and range. (3 marks)

23. Functions, 2ADV F2 SM-Bank 3

The diagram below shows part of the graph of the function with rule

$$f(x) = k \log_e(x + a) + c, \text{ where } k, a \text{ and } c \text{ are real constants.}$$

- The graph has a vertical asymptote with equation $x = -1$.
- The graph has a y-axis intercept at 1.
- The point P on the graph has coordinates $(p, 10)$, where p is another real constant.



- State the value of a . (1 mark)
- Find the value of c . (1 mark)
- Show that $k = \frac{9}{\log_e(p + 1)}$. (2 marks)

24. Functions, 2ADV F2 SM-Bank 10

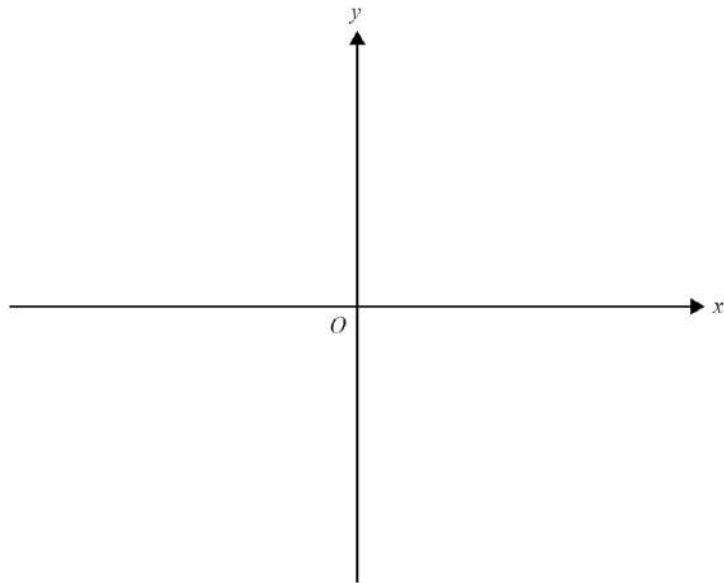
Consider the function $f(x) = \frac{x}{4 - x^2}$.

- Identify the domain of $f(x)$. (1 mark)
- Sketch the graph $y = f(x)$, showing all intercepts and asymptotes. (3 marks)

25. Functions, 2ADV F2 EQ-Bank 11

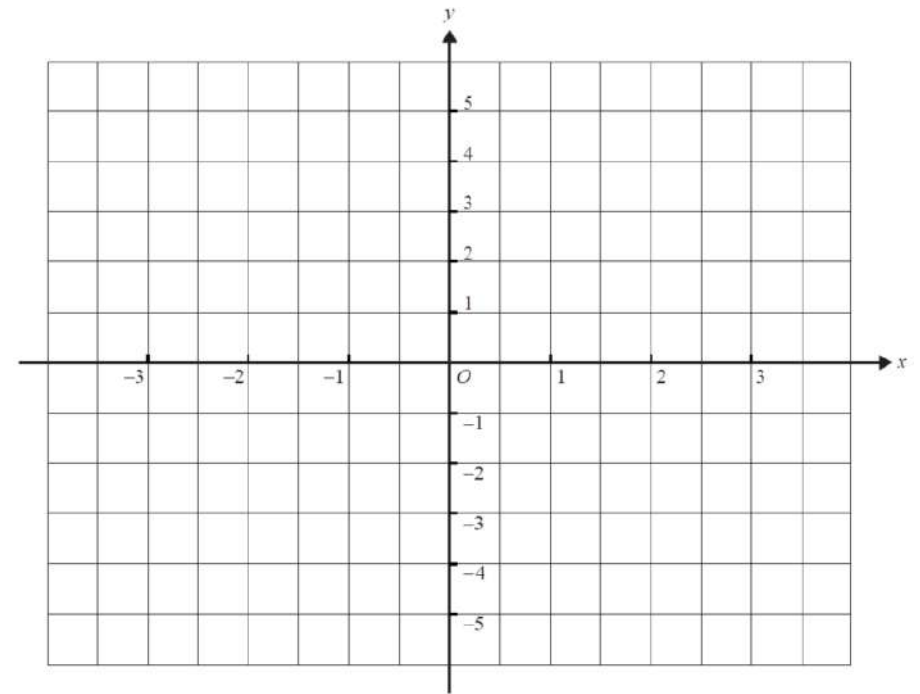
On the axes below, sketch the graph of $f(x) = \frac{2x-2}{x+1}$.

Label all axis intercepts. Label each asymptote with its equation. (4 marks)



26. Functions, 2ADV F2 SM-Bank 12

Sketch the graph of $f(x) = 2 + \frac{3}{x-1}$. Label the axis intercepts with their coordinates and label any asymptotes with the appropriate equation. (4 marks)



27. Functions, 2ADV' F2 2007 HSC 3b

(i) Find the vertical and horizontal asymptotes of the hyperbola

$y = \frac{x-2}{x-4}$ and hence sketch the graph of $y = \frac{x-2}{x-4}$. (3 marks)

(ii) Hence, or otherwise, find the values of x for which

$$\frac{x-2}{x-4} \leq 3. \text{ (2 marks)}$$

28. Functions, 2ADV F2 EQ-Bank 13

The curve $y = kx^2 + c$ is subject to the following transformations

- Translated 2 units in the positive x -direction
- Dilated in the positive y -direction by a factor of 4
- Reflected in the y -axis

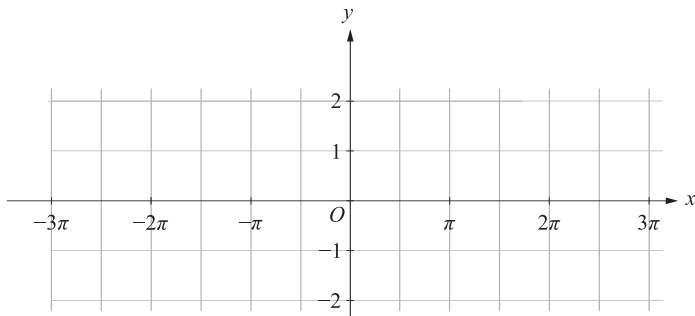
The final equation of the curve is $y = 8x^2 + 32x - 8$.

- Find the equation of the graph after the dilation. (1 mark)
- Find the values of k and c . (2 marks)

29. Functions, 2ADV F2 EQ-Bank 3

By drawing graphs on the number plane, show how many solutions exist for the equation

$$\cos x = \left| \frac{x - \pi}{4} \right| \text{ in the domain } (-\infty, \infty) \text{ (3 marks)}$$



30. Functions, 2ADV F2 2013 HSC 15c

- Sketch the graph $y = |2x - 3|$. (1 mark)
- Using the graph from part (i), or otherwise, find all values of m for which the equation $|2x - 3| = mx + 1$ has exactly one solution. (2 marks)

Worked Solutions

1. Functions, 2ADV F2 2016 HSC 3 MC

$$y = 3 - (x - 2)^2$$

Curve is concave down and passes through (2, 3)

$$\Rightarrow B$$

2. Functions, 2ADV F2 EQ-Bank 2 MC

Transforming $y = x^2$:

Translate 1 unit left $\Rightarrow y = (x + 1)^2$

Dilate from x -axis by a factor of 2 $\Rightarrow y = 2(x + 1)^2$

Reflect in x -axis $\Rightarrow y = -2(x + 1)^2$

Translate 1 unit up $\Rightarrow y = 1 - 2(x + 1)^2$

$\therefore$ Transformations describe graph A.

$$\Rightarrow A$$

3. Functions, 2ADV' F2 2019 HSC 4 MC

By elimination:

Graph is an even function

$$\Rightarrow f(x) = f(-x)$$

$\therefore$ Eliminate A and C

When $-1 < x < 1$, $y \leq 0$

$\therefore$ Eliminate D

$$\Rightarrow B$$

4. Functions, 2ADV F2 SM-Bank 5 MC

1st transformation:

$$P(4, -3) \rightarrow (4, 1)$$

2nd transformation:

$$(4, 1) \rightarrow (-4, 1)$$

$$\Rightarrow A$$

5. Functions, 2ADV F2 2014 HSC 2 MC

$$y = (x - 1)^2 \rightarrow \text{vertex } (1, 0) \\ \Rightarrow B$$

6. Functions, 2ADV F2 SM-Bank 6 MC

Let $y = \sqrt{2x - 5}$

1st transformation:

$$y = -\sqrt{2x - 5}$$

2nd transformation:

$$y = -\sqrt{2(2x) - 5} \\ = -\sqrt{4x - 5} \\ \therefore f(x) = -\sqrt{4x - 5} \\ \Rightarrow D$$

7. Functions, 2ADV F2 SM-Bank 4 MC

Let $y = 3x^{\frac{5}{2}}$

Reflect in the x -axis:

$$y = -3x^{\frac{5}{2}}$$

Translate 3 units to the right:

$$y = -3(x - 3)^{\frac{5}{2}}$$

Translate 4 units down:

$$y = -3(x - 3)^{\frac{5}{2}} - 4$$

$$\Rightarrow B$$

8. Functions, 2ADV F2 SM-Bank 10 MC

$$y = -\left(\frac{3 - x}{2 - x}\right) \\ = -\left(\frac{2 - x}{2 - x} + \frac{1}{2 - x}\right) \\ = -1 - \frac{1}{2 - x} \\ \therefore \text{Asymptotes: } x = 2, y = -1 \\ \Rightarrow C$$

9. Functions, 2ADV' F2 2015 HSC 5 MC

$$y = \frac{3x}{(x + 1)(x + 2)} \\ \text{Asymptotes at } x = -1 \text{ and } x = -2 \\ \text{As } x \rightarrow \infty, y \rightarrow 0^+ \\ \text{As } x \rightarrow -\infty, y \rightarrow 0^- \\ \therefore \text{Horizontal asymptote at } y = 0 \\ \Rightarrow A$$

10. Functions, 2ADV' F2 2017 HSC 5 MC

$$y = \frac{2x^2}{(1 - x^2)} \\ = -\frac{(2 - 2x^2 - 2)}{(1 - x^2)} \\ = -\frac{2(1 - x^2)}{(1 - x^2)} - \frac{2}{(1 - x^2)} \\ = -2 - \frac{2}{(1 - x^2)} \\ \text{As } x \rightarrow \infty, y \rightarrow -2 \\ \therefore \text{Horizontal asymptote at } y = -2 \\ \Rightarrow D$$

11. Functions, 2ADV F2 SM-Bank 7 MC

$$g(x) = \frac{1}{2}f(x-1), A(3, 2)$$

Dilate by a factor of $\frac{1}{2}$ from x -axis:

$$A(3, 2) \rightarrow A'(3, 1)$$

Translate 1 unit to right:

$$A'(3, 1) \rightarrow P(4, 1)$$

$\Rightarrow C$

12. Functions, 2ADV F2 SM-Bank 8 MC

$$\text{Let } f(x) = \sqrt{8x^3 + 1}$$

$$\begin{aligned} f\left(\frac{1}{2}x\right) &= \sqrt{8\left(\frac{1}{2}x\right)^3 + 1} \\ &= \sqrt{x^3 + 1} \end{aligned}$$

$$f\left(\frac{1}{2}x\right) \rightarrow \text{dilate by factor of 2 from } y\text{-axis}$$

$\Rightarrow A$

13. Functions, 2ADV F2 EQ-Bank 5

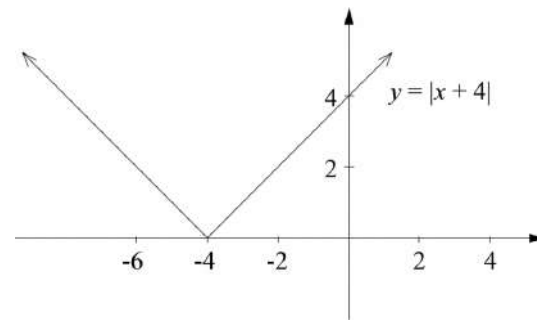
$$g(x) = \frac{1}{3}\sin(4x)$$

$\Rightarrow$ The new amplitude is one third of the original amplitude.

$$\text{Period} = \frac{2\pi}{n} \Rightarrow n = \frac{1}{4}$$

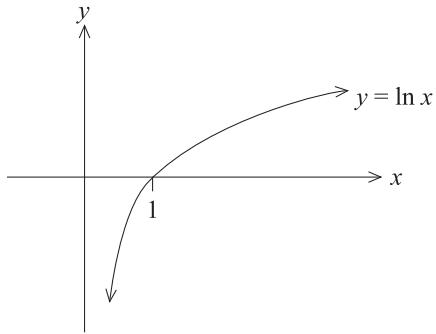
$\Rightarrow$ The new period is one quarter of the original period.

14. Functions, 2ADV F2 2006 HSC 1c



15. Functions, 2ADV F2 SM-Bank 1

i.

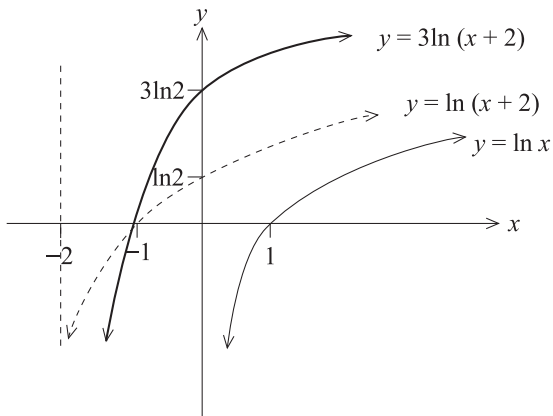


ii. Transforming $y = \ln x \Rightarrow y = \ln(x + 2)$

$y = \ln x \Rightarrow$ shift 2 units to left.

Transforming $y = \ln(x + 2)$ to $y = 3\ln(x + 2)$

$\Rightarrow$ increase each y value by a factor of 3



16. Functions, 2ADV F2 EQ-Bank 9

i. Transformation 1:

Translate $y = x^4$ 2 units to the left.

$$y = x^4 \Rightarrow y = (x + 2)^4$$

Transformation 2:

Dilate $y = (x + 2)^4$ by a factor of $\frac{1}{3}$ from the x -axis

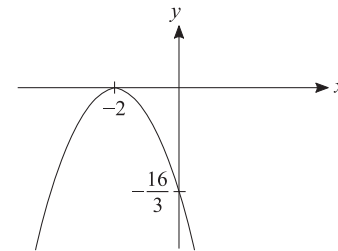
$$y = (x + 2)^4 \Rightarrow y = \frac{(x + 2)^4}{3}$$

Transformation 3:

Reflect $y = \frac{(x + 2)^4}{3}$ in the x -axis.

$$y = \frac{(x + 2)^4}{3} \Rightarrow y = -\frac{(x + 2)^4}{3}$$

ii.



17. Functions, 2ADV F2 EQ-Bank 9

i. $4x - 1 \neq 0$

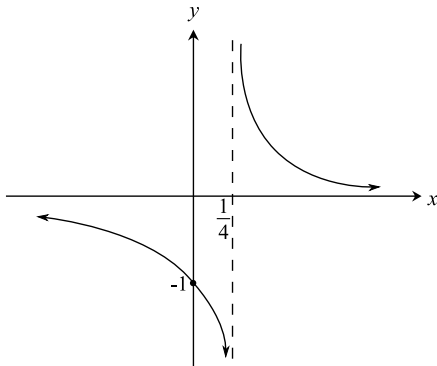
$$x \neq \frac{1}{4}$$

$$\therefore \text{Domain: } \left\{ \text{all real } x, x \neq \frac{1}{4} \right\}$$

ii. When $x = 0, y = -1$

$$\text{As } x \rightarrow \infty, y \rightarrow 0^+$$

$$\text{As } x \rightarrow -\infty, y \rightarrow 0^-$$



18. Functions, 2ADV' F2 2012 HSC 13b

$$\begin{aligned} \text{(i) } y &= \frac{2x^2}{x^2 + 9} \\ &= \frac{2}{1 + \frac{9}{x^2}} \end{aligned}$$

$$\text{As } x \rightarrow \infty, y \rightarrow 2$$

$$\text{As } x \rightarrow -\infty, y \rightarrow 2$$

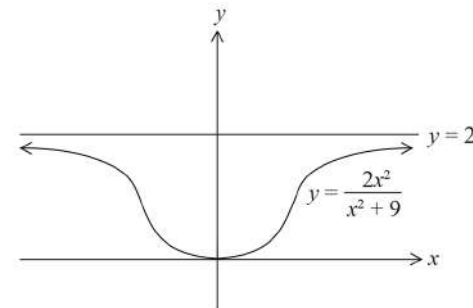
$$\therefore \text{Horizontal asymptote at } y = 2$$

(ii) At $x = 0, y = 0$

$$f(x) = \frac{2x^2}{x^2 + 9} \geq 0 \text{ for all } x$$

$$f(-x) = \frac{2(-x)^2}{(-x)^2 + 9} = \frac{2x^2}{x^2 + 9} = f(x)$$

$$\text{Since } f(x) = f(-x) \Rightarrow \text{EVEN function}$$



19. Functions, 2ADV F2 EQ-Bank 1

$$y = |x|$$

Translate 3 units left $\Rightarrow y = |x + 3|$

Reflect in the x -axis $\Rightarrow y = -|x + 3|$

Multiply by $\frac{1}{3} \Rightarrow y = -\frac{1}{3}|x + 3|$

Translate 2 units up $\Rightarrow y = -|x + 3| + 2$

$$\therefore k = -\frac{1}{3}, b = 3, c = 2$$

20. Functions, 2ADV F2 EQ-Bank 14

$$y = x^2$$

Transformation 1:

Translate 3 units in negative x -direction

$$y = (x + 3)^2$$

$$y = x^2 + 6x + 9$$

Transformation 2:

Translate 10 units in negative y -direction

$$y = x^2 + 6x - 1$$

Transformation 3:

Reflect in the x -axis

$$y = -(x^2 + 6x - 1)$$

$$= 1 - 6x - x^2$$

21. Functions, 2ADV F2 SM-Bank 13

i. $f(x) = \frac{1 - e^x}{1 + e^x}$

$$\begin{aligned} f(-x) &= \frac{1 - e^{-x}}{1 + e^{-x}} \times \frac{e^x}{e^x} \\ &= \frac{e^x - 1}{e^x + 1} \\ &= -\frac{1 - e^x}{1 + e^x} \\ &= -f(x) \end{aligned}$$

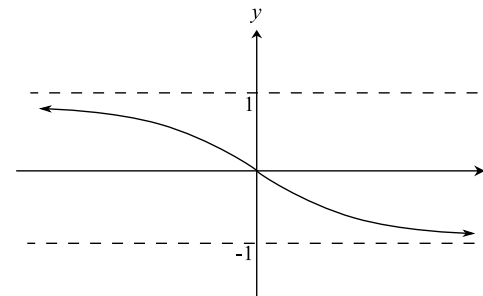
$\therefore f(x)$ is ODD.

ii. $y = \frac{1 - e^x}{1 + e^x} \times \frac{e^{-x}}{e^{-x}} = \frac{e^{-x} - 1}{e^{-x} + 1} = 1 - \frac{2}{e^{-x} + 1}$

As $x \rightarrow \infty$, $\frac{2}{e^{-x} + 1} \rightarrow 2$, $y \rightarrow -1$

As $x \rightarrow -\infty$, $\frac{2}{e^{-x} + 1} \rightarrow 0$, $y \rightarrow 1$

When $x = 0$, $y = 0$

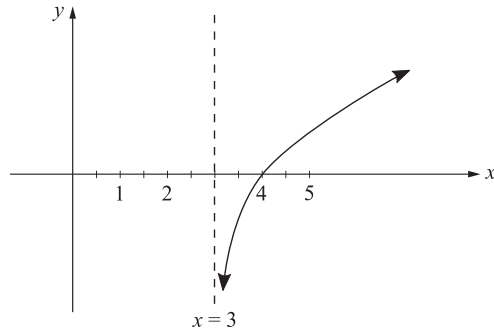


22. Functions, 2ADV F2 SM-Bank 2

Asymptote when $x = 3$

Domain $\{x: x > 3\}$

Range: all y



23. Functions, 2ADV F2 SM-Bank 3

i. Vertical Asymptote:

$$x = -1$$

$$\therefore a = 1$$

ii. Solve $f(0) = 1$ for c ,

$$c = 1$$

iii. $f(x) = k \log_e(x + 1) + 1$

Since $f(p) = 10$,

$$k \log_e(p + 1) + 1 = 10$$

$$k \log_e(p + 1) = 9$$

$$\therefore k = \frac{9}{\log_e(p + 1)} \dots \text{as required}$$

24. Functions, 2ADV F2 SM-Bank 10

$$(i) 4 - x^2 \neq 0$$

$$\Rightarrow x \neq 2 \text{ or } -2$$

$$\therefore \text{Domain: all } x, x \neq \pm 2$$

(ii) Asymptotes at $x = \pm 2$

Passes through $(0, 0)$

$$\text{As } x \rightarrow 2(-), y \rightarrow \infty$$

$$x \rightarrow 2(+), y \rightarrow -\infty$$

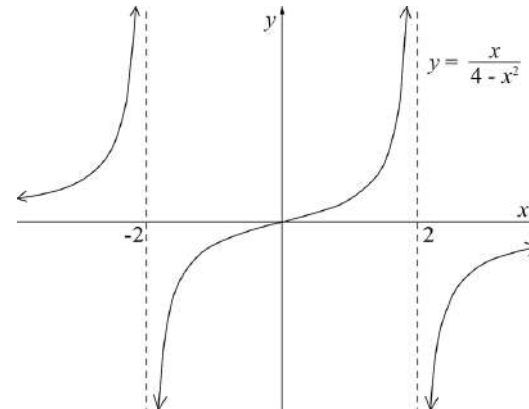
$$\text{As } x \rightarrow -2(-), y \rightarrow \infty$$

$$x \rightarrow -2(+), y \rightarrow -\infty$$

$$\text{As } x \rightarrow \infty, y \rightarrow 0$$

$$x \rightarrow -\infty, y \rightarrow 0$$

COMMENT: Note that $x \rightarrow 2(-)$ is a short way of writing as x approaches 2 from the negative (or left-hand) side. This notation can save time when required.



25. Functions, 2ADV F2 EQ-Bank 11

$$\begin{aligned}\frac{2x-2}{x+1} &= \frac{2(x+1)-4}{x+1} \\ &= 2 - \frac{4}{(x+1)}\end{aligned}$$

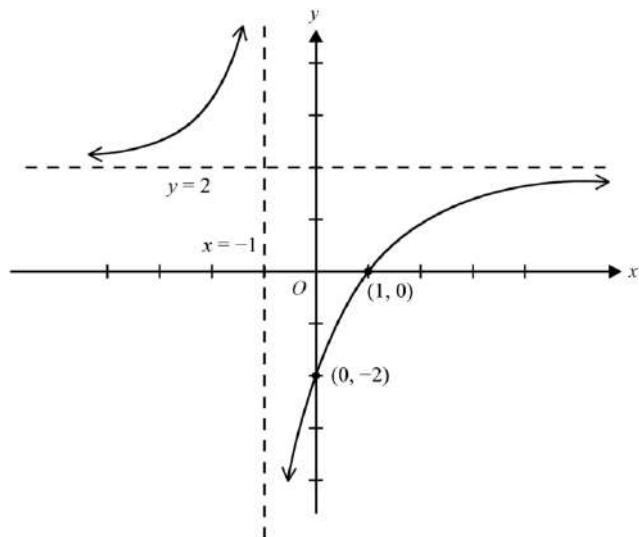
Asymptotes: $x = -1$, $y = 2$

As $x \rightarrow \infty$, $y \rightarrow 2(-)$

As $x \rightarrow -\infty$, $y \rightarrow 2(+)$

As $x \rightarrow -1(-)$, $y \rightarrow \infty$

As $x \rightarrow -1(+)$, $y \rightarrow -\infty$



26. Functions, 2ADV F2 SM-Bank 12

$$\begin{aligned}\frac{2x+1}{x-1} &= \frac{2x-2+3}{x-1} \\ &= 2 + \frac{3}{x-1}\end{aligned}$$

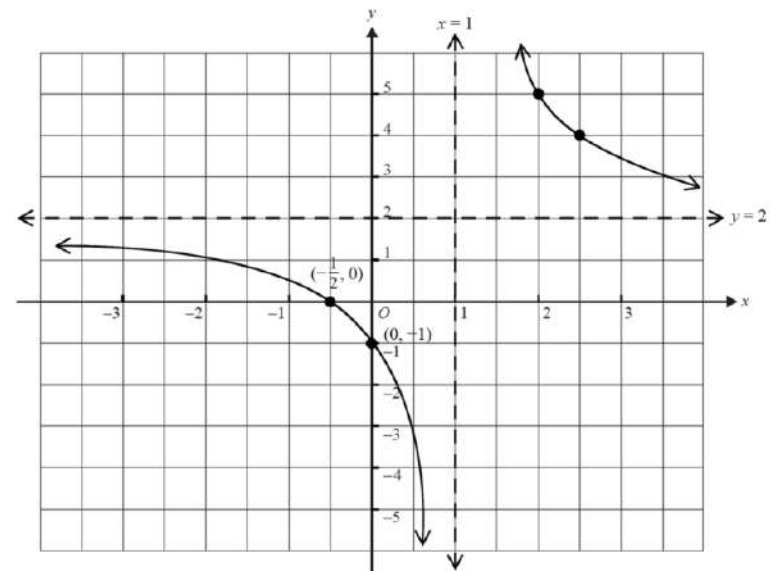
Asymptotes: $x = 1$, $y = 2$

As $x \rightarrow \infty$, $y \rightarrow 2(+)$

As $x \rightarrow -\infty$, $y \rightarrow 2(-)$

As $x \rightarrow 1(-)$, $y \rightarrow -\infty$

As $x \rightarrow 1(+)$, $y \rightarrow \infty$



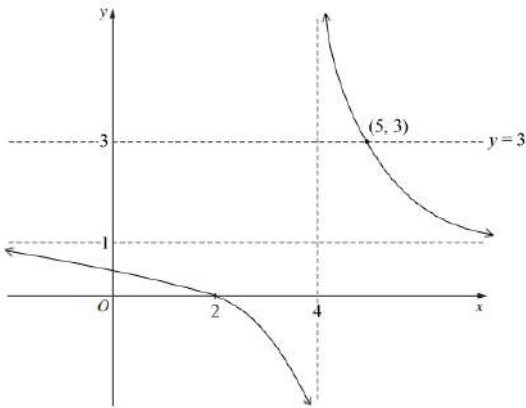
(i) $y = \frac{x-2}{x-4}$

Vertical asymptote at $x = 4$

$$\lim_{x \rightarrow \infty} \frac{x-2}{x-4} = \lim_{x \rightarrow \infty} \frac{1 - \frac{2}{x}}{1 - \frac{4}{x}} = 1$$

$$y\text{-intercept} = \frac{1}{2}$$

$$x\text{-intercept} = 2$$



(ii) Find x so that $\frac{x-2}{x-4} \leq 3$

When $\frac{x-2}{x-4} = 3$

$$x-2 = 3x-12$$

$$2x = 10$$

$$x = 5$$

$$\Rightarrow (5, 3) \text{ is the intersection of } y = 3 \text{ and } y = \frac{x-2}{x-4}$$

$$\therefore \frac{x-2}{x-4} \leq 3 \text{ when } x < 4 \text{ and } x \geq 5.$$

i. $y = kx^2 + c$

Translate 2 units in positive x -direction.

$$y = kx^2 + c \Rightarrow y = k(x-2)^2 + c$$

Dilate in the positive y -direction by a factor of 4.

$$y = k(x-2)^2 + c \Rightarrow y = 4k(x-2)^2 + 4c$$

ii. $y = 4k(x^2 - 4x + 4) + 4c$

$$= 4kx^2 - 16kx + 16k + 4c$$

Reflect in the y -axis:

$$y = 4k(-x)^2 - 16k(-x) + 16k + 4c$$

$$= 4kx^2 + 16kx + 16k + 4c$$

Equating co-efficients:

$$4k = 8$$

$$\therefore k = 2$$

$$16k + 4c = -8$$

$$4c = -40$$

$$\therefore c = -10$$

29. Functions, 2ADV F2 EQ-Bank 3

Sketch:

$$y = \cos x$$

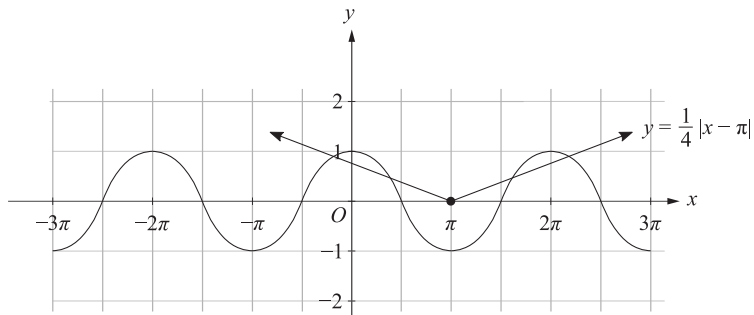
$$y = \left| \frac{x - \pi}{4} \right|$$

Translate π units to the right:

$$y = |x| \Rightarrow y = |x - \pi|$$

Multiply by $\frac{1}{4}$:

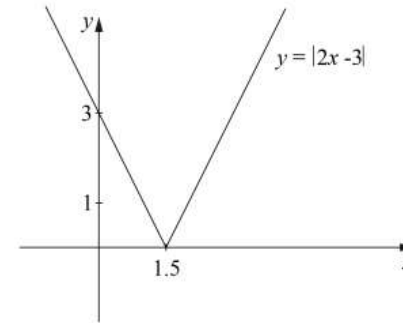
$$y = |x - \pi| \Rightarrow y = \frac{1}{4}|x - \pi| = \left| \frac{x - \pi}{4} \right|$$



$\therefore$ There are 4 solutions.

30. Functions, 2ADV F2 2013 HSC 15c

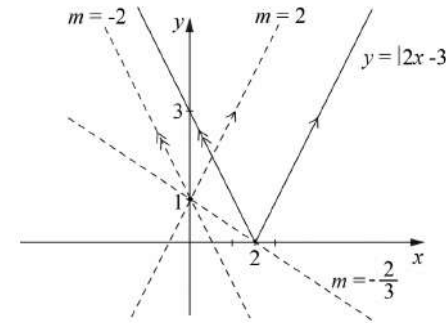
i.



♦ Mean mark 49%

MARKER'S COMMENT: Many students drew diagrams that were "too small", didn't use rulers or didn't use a consistent scale on the axes!

ii.



Line of intersection $y = mx + 1$ passes through $(0, 1)$

If it also passes through $(1.5, 0) \Rightarrow 1$ solution

$$\begin{aligned} m &= \frac{y_2 - y_1}{x_2 - x_1} \\ &= \frac{1 - 0}{0 - \frac{3}{2}} \\ &= -\frac{2}{3} \end{aligned}$$

♦♦ Mean mark 25%.

COMMENT: Students need a clear graphical understanding of what they are finding to solve this very challenging, Band 6 question.

Gradients of $y = |2x - 3|$ are 2 or -2

Considering a line through $(0, 1)$: Similarly,

If $m < -2$, only intersects once.

If $m \geq 2$, only intersects once.

$\therefore$ Only one solution when $m = -\frac{2}{3}$, $m \geq 2$ or $m < -2$